

Origami konstrukcije in reševanje problemov s prepogibanjem papirja

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- 4 Reševanje enačb
- 5 Alhazenov problem

Osnovna literatura

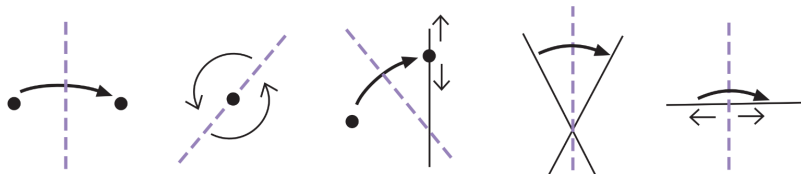
1. Hull, T. C., *Origametry: Mathematical Methods in Paper Folding*, Cambridge University Press, 2020.
2. Martin, G. E., *Geometric Constructions*, Springer New York, 1997.
3. Geretschläger, R., *Euclidean Constructions and the Geometry of Origami*, Mathematics Magazine **68**(5) (1995) 357–371, dostopno na <http://www.jstor.org/stable/2690924>.
4. Haga, K., *ORIGAMICS: Mathematical Explorations Through Paper Folding*, World Scientific Publishing Co. Pte. Ltd., 2008.
5. Nishimura, Y., *Solving Alhazen's problem by Origami*, International Journal of Geometry **7**(2) (2018) 37–42.

Origami operacije

Pravila prepogibanja:

- enkratni in ravni pregibi
- dodatna orodja

Vsi možni pregibi:



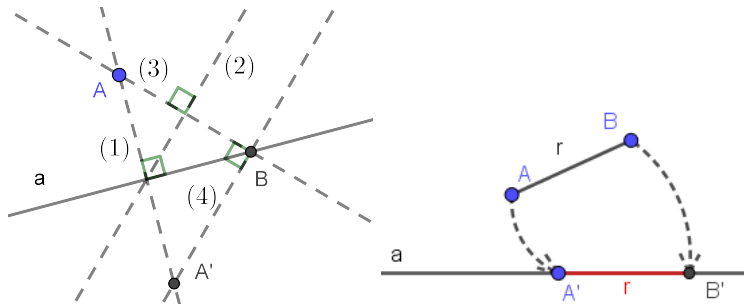
Origami operacije O1–O8.

Izrek 2.7

Če dovolimo le enkratne in ravne pregibe, so edine možne operacije prepogibanja operacije O1–O8.

Trditvi 2.10 in 2.11

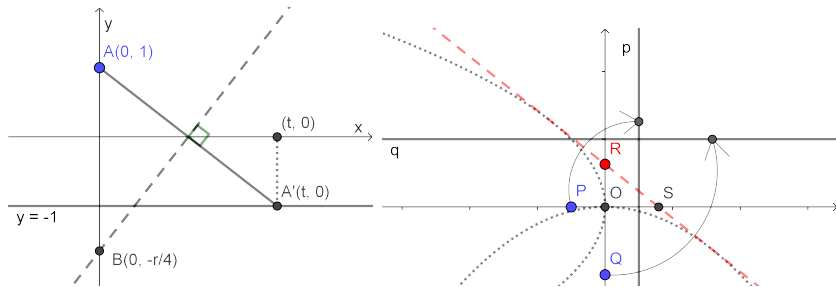
Z ravnimi in enkratnimi prepogibi ter upoštevanjem origami operacij lahko s prepogibanjem papirja zrcalimo točke in prenašamo razdalje.



$\mathbb{O}$: množica origami števil

Izrek 2.29 in trditev 2.30

Množica $\mathbb{O}$ je podpolje polja $\mathbb{C}$. Za $\alpha \in \mathbb{O}$ velja $\sqrt{\alpha}, \sqrt[3]{\alpha} \in \mathbb{O}$.



Posledica 2.31

Množica evklidsko-konstruktibilnih števil je prava podmnožica množice $\mathbb{O}$.

Origami števila so tista števila, ki jih lahko zapišemo z znaki

$$() 0 1 2 3 4 5 6 7 8 9 i + - \times \div \sqrt{} \sqrt[3]{}.$$

Konstrukcija pravih n -kotnikov

Gauss-Wantzlov izrek (izrek 2.38)

Pravilni n -kotnik je evklidsko-konstruktibilen natanko takrat, ko je $n = 2^a p_1 \cdots p_k$ za neka $a, k \in \mathbb{N}_0$, kjer so $p_1, \dots, p_k$ različna Fermatova praštevila, tj. $p_i = 2^{2^{m_i}} + 1$ kjer je $m_i \in \mathbb{N}_0$.

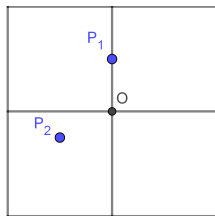
3, 4, 5, 6, 8, 10, 12, 15, 16, 17, 20

Izrek 2.40

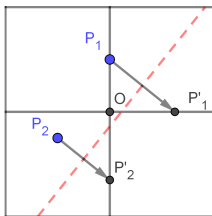
Pravilni n -kotnik se lahko konstruira z origamijem natanko takrat, ko je $n = 2^a 3^b p_1 \cdots p_k$ za neke $a, b, k \in \mathbb{N}_0$, kjer so $p_1, \dots, p_k$ različna Pierpontova praštevila, tj. $p_i > 3$ in $p_i = 2^{m_i} 3^{n_i} + 1$ za neka $m_i, n_i \in \mathbb{N}_0$.

3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18, 19, 20

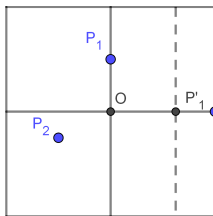
Konstrukcija pravega 7-kotnika



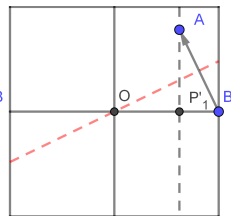
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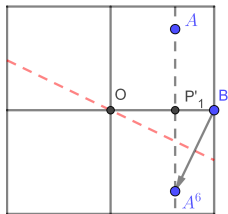
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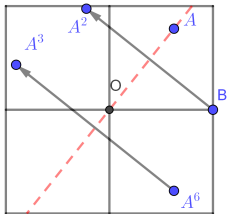
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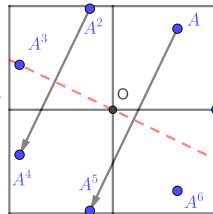
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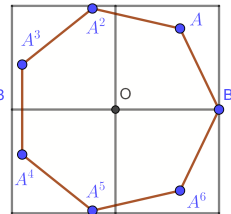
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6.

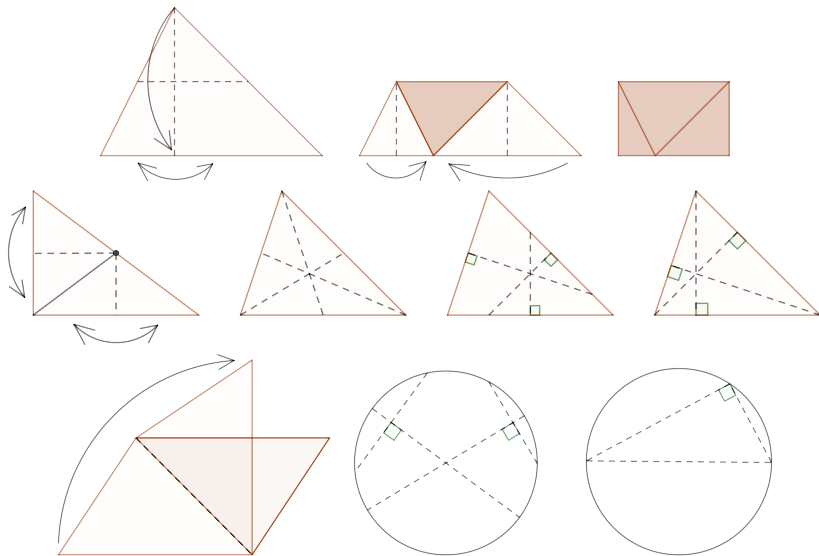


7.

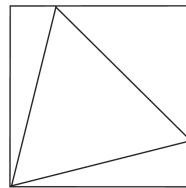
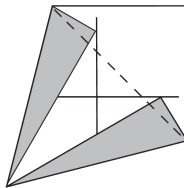
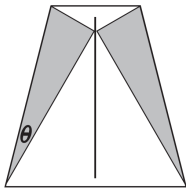
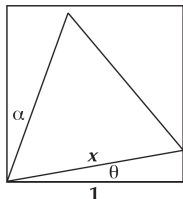
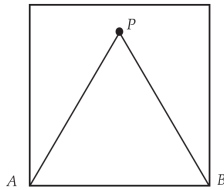
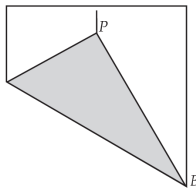
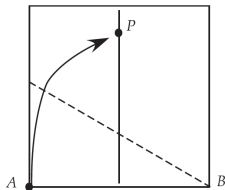
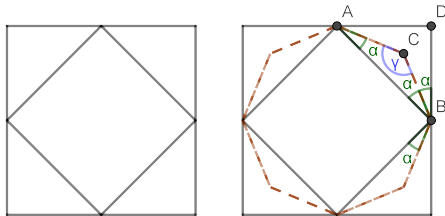


8.

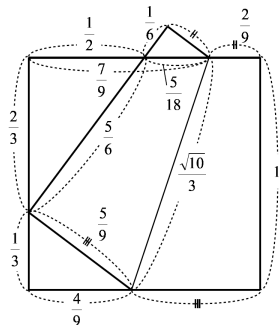
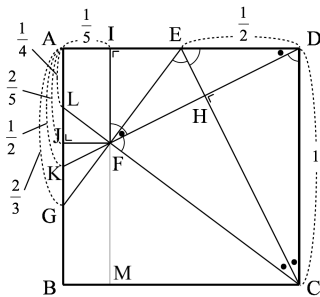
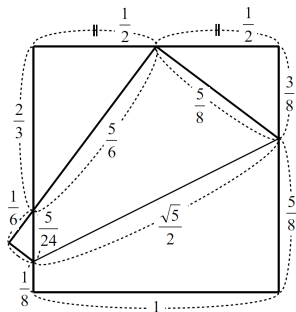
Uporaba v osnovni šoli



Konstrukcije nekaterih pravilnih likov



Hagovi izreki



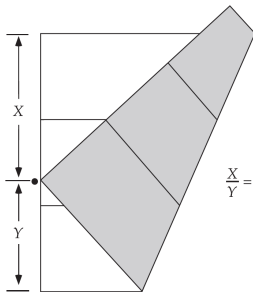
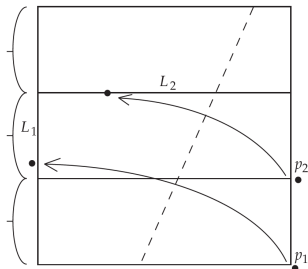
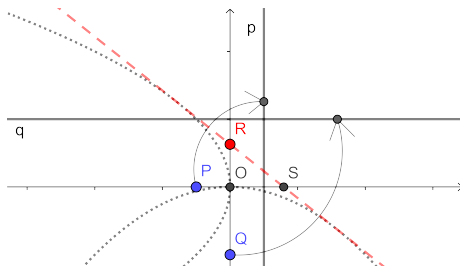
Razdelicev daljice na n delov

X-pregibi

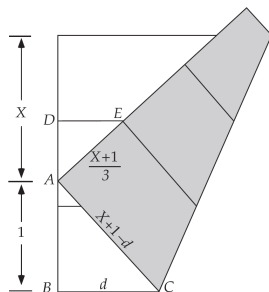
Starogrški problemi

- **Podvojitev kocke:** Za dano kocko konstruiraj novo kocko, ki ima dvakrat večji volumen od prve (konstrukcija razdalje $\sqrt[3]{2}$).
- **Trisekcija kota:** Za dan kot α konstruiraj kot $\alpha/3$.
- **Kvadratura kroga:** Za dan krog konstruiraj kvadrat, ki ima enako ploščino kot dani krog (konstrukcija razdalje $\sqrt{\pi}$).

Podvojitev kocke – Martinova in Messerjeva konstrukcija



$$\frac{X}{Y} = \sqrt[3]{2}$$



Trisekcija kota

Tangente na stožnice

